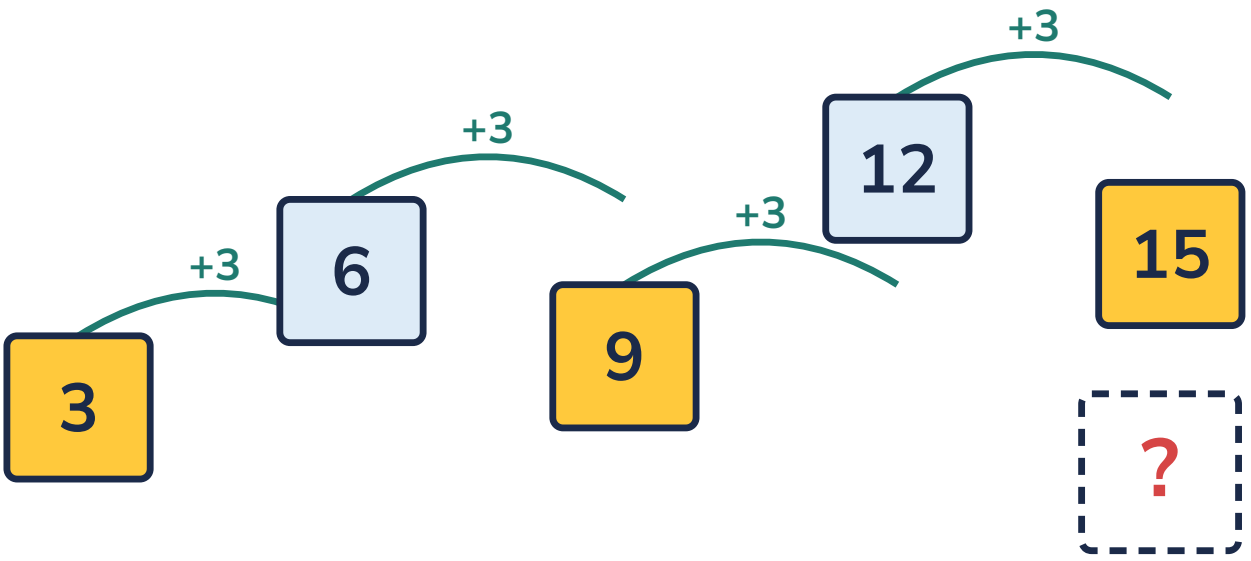
FREE TASTER • FOR PARENTS AND PRIMARY 1 CHILDREN

Think Like a Mathematician

Six puzzles to try at home. About 10 minutes each. No shopping needed.



Notice it. Test it. Explain why.

From the Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

00 Four rules for every puzzle

1 Notice it

What repeats? What changes?

2 Test it

Try it with coins, buttons or spoons.

3 Check it

Does your answer make sense?

4 Explain why

Say how you know.

How to help (for grown-ups)

1. **Ask before you tell.** Start with "What do you notice?"
2. **Use real things.** Coins, buttons, spoons or pasta make numbers easy to see.
3. **Stuck?** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right.
5. **Praise the thinking,** not the cleverness: "I like how you checked that."
6. **Keep it short.** Stop while it is still fun.

> **Hard question? Great! Mathematicians get stuck all the time. That is when thinking happens.**

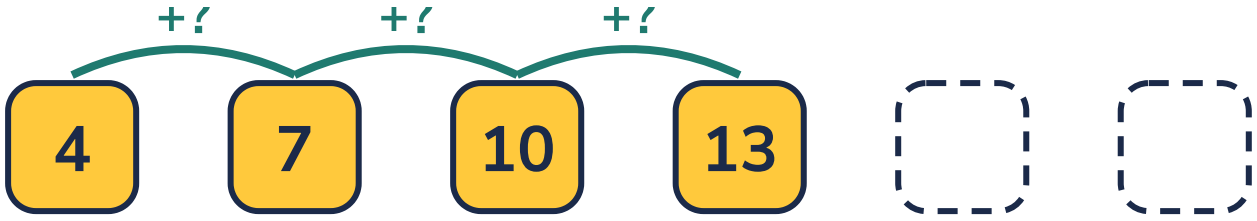
01 Frog Jumps

PUZZLE SET 1 • PATTERNS

Find the rule, then write the next two numbers. Try it with a number line, or hop a coin along the table.

Start here

1 4, 7, 10, 13, __, __ What is the rule?



The rule is: _____

2 5, 10, 15, 20, __, __



The rule is: _____

3 1, 5, 9, 13, __, __



The rule is: _____

01 Frog Jumps: think harder

PUZZLE SET 1 • PATTERNS

Think harder

4 3, 8, __, 18, __, 28

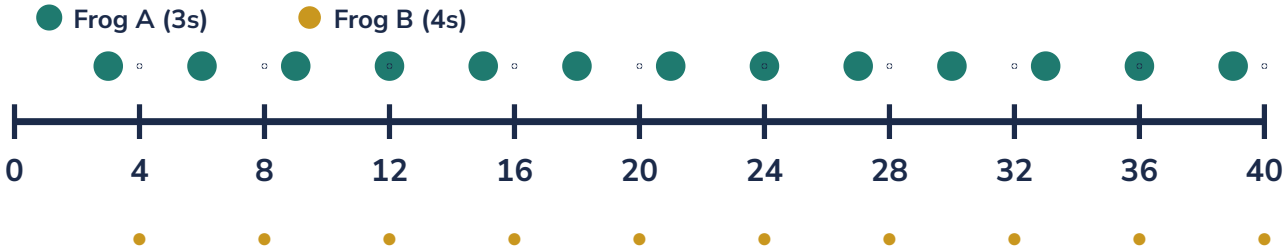


5 40, __, 28, __, 16, 10



Go further (only if it is fun)

6 **Frog A** jumps in 3s from 0. **Frog B** jumps in 4s from 0. Which numbers up to 40 do **both** frogs land on? (Do not count 0.)



Explain why: how do you know you found them all?

Try it with objects

Line up 40 pasta pieces. Put a coin on every 3rd piece, and a button on every 4th. Where do the coin and the button sit together?

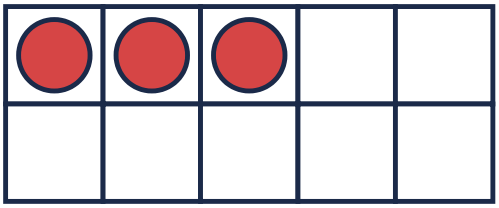
02 Making 10

PUZZLE SET 2 • NUMBER BONDS

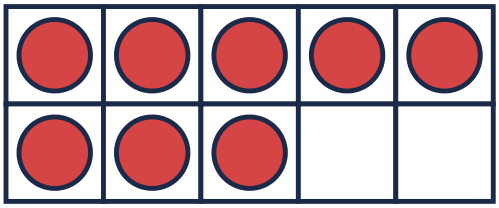
Put 10 buttons or counters in two groups to help you.

Start here

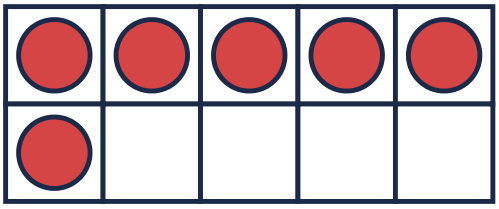
1 10 = 3 + __



2 10 = __ + 8

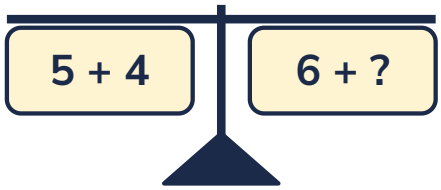


3 10 = 6 + __



Think harder: keep it balanced

4 5 + 4 = 6 + __



5 8 + 2 = 7 + __

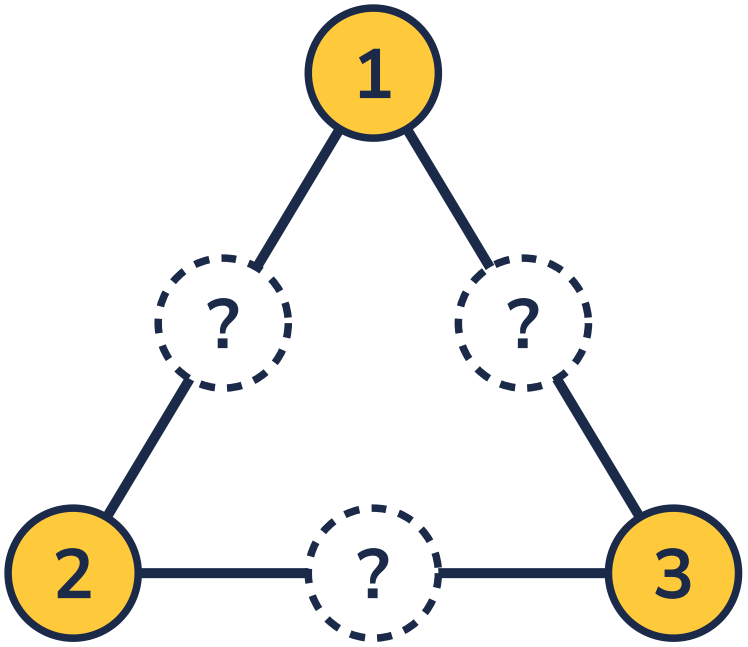
Both sides must be the same. Check by working out each side.

02 The magic triangle

PUZZLE SET 2 • NUMBER BONDS

Go further

6 Put **1, 2, 3, 4, 5, 6** in the spaces so that **each side adds to 9**. The corners are 1, 2 and 3. What goes in the middle of each side?



Explain why: how do you know each side adds to 9?

Try it with objects

Write 1 to 6 on six bits of paper. Move them around until every side makes 9.

LOOK ONLY AFTER YOU HAVE TRIED

Answers

A different way of thinking is fine, as long as you can explain it.

Set 1: Frog Jumps

1. 16, 19. Add 3.
2. 25, 30. Add 5.
3. 17, 21. Add 4.
4. 13 and 23. Add 5.
5. 34 and 22. Take away 6.
6. 12, 24 and 36. Both frogs land there because we say 12 when counting in 3s and in 4s. After that it happens every 12.

Set 2: Making 10

1. 7
2. 2
3. 4
4. 3 (5 + 4 = 9, 6 + 3 = 9)
5. 3 (8 + 2 = 10, 7 + 3 = 10)
6. Between 1 and 2: 6. Between 2 and 3: 4. Between 3 and 1: 5. Each side adds to 9.

KEEP THINKING

Want more?

Weekly Dose of Thinking Math

Join our free list. Each week you get a new set of puzzles and a five-minute "explain why" question for the dinner table.

[Sign-up link or QR code]

Math Inquirers Holiday Program

Ten 90-minute lessons over five weeks, live online in small groups of up to 8 children. Hands-on puzzles, a printed workbook posted to your home with a cut-out kit, a stamped Inquiry Journal and a certificate.

[Dates, fee and sign-up link]

Questions?

[Name, email or phone]

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